SIMATS ENGINEERING

ASSESSMENT TOOL -1

NAME: M.SADHANA

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COURSE: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA04

DATE: 3/10/2026

COURSE OUTCOME COVERED.

CO1: Analyse the role of data science, facts of data and the big data ecosystem for informed decision making.

CODE OF CONDUCT.

I am M. Sadhana (1924260086), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action

Student signature: M. Sadhana.

1. Identifying Students who may fail :-

* Analyze marks, attendance and previous performance.
* Predict Students who are at risk
* Provide early support to improve their results.

2. Hospital patient records :-

* Analyze historical patient data.
* Identify disease patterns and risks
* Help doctors make better decisions

3. Online product ~~product~~ recommendations :-

* uses a recommendation system
* Analyzes customer searches and purchases.
* Suggest products based on customer interests.

4. Festival product sales :-

* Analyze previous festival sales.
* Identify high demand products.
* Helps maintain sufficient stock

5. Improving crop yield :-

* Analyse soil and weather conditions
* Predict suitable crops and farming conditions.
* Helps farmers improve productivity.

6. Weather updating every minute :-

* This shows Velocity.
* Data is generated and updated quickly.
* It requires real time data processing

7. Social media customer comments :-

* These are mainly unstructured data.
* Comments are usually text.
* They can be analyzed for opinions and sentiments.

8. Missing ages and Incorrect Phone numbers :-

* This is a data quality problem.
* It involves missing and inaccurate data.
* Data cleaning is required.

9. Storing large files in Hadoop :-

* HDFS is used.
* It stores large datasets across multiple computers.
* It provides distributed data.

10. Collecting live IoT sensor data :-

* Apache kafka can be used.
* It collects and transfers real time da.
* It is useful for streaming data.